✅ Month 5 - Project Title: Flight Delay Prediction Using Machine Learning

👨‍💼 Client Overview

An airline company wants to predict delays in flights using weather, airline, and route data to improve customer satisfaction.

🎯 Project Objective

Train a machine learning classification model to predict whether a flight will be delayed or on time.

📦 Dataset

Airline Delay Dataset (Kaggle)

🔧 Step-by-Step Project Requirements

1. Data Preparation

- Drop columns with too many missing values
- Encode Airline, Origin, Dest with LabelEncoder or OneHot
- Convert FlightDate to weekday, month

2. Model Building

- Train/Test Split
- Use Logistic Regression & Random Forest
- Evaluate with Accuracy, Precision, Recall

3. Evaluation

- Confusion Matrix
- ROC Curve
- Classification Report

4. Deliverables

- .ipynb notebook with trained model
- Saved .pkl model file
- Model evaluation report